

23- S1- Q3

$$Q: 1000 \begin{cases} 600 & : 1 \\ 400 & : 2 \end{cases}$$

$$m_1 = \begin{bmatrix} 0.023 \\ 0.061 \end{bmatrix} \quad m_2 = \begin{bmatrix} 1.45 \\ 1.98 \end{bmatrix}$$

$$S_1 = \begin{bmatrix} 615.87 & 55.83 \\ 55.83 & 539.64 \end{bmatrix} \quad S_2 = \begin{bmatrix} 426.49 & -1.77 \\ -1.77 & 387.20 \end{bmatrix}$$

(a) DLA

$$x_1 = \begin{bmatrix} 1.52 \\ -1.61 \end{bmatrix}$$

$$(b) p(x) = \frac{1}{\sqrt{2\pi} \sigma} e^{\left[-\frac{1}{2} \left(\frac{x-\mu}{\sigma}\right)^2\right]}$$

$$NBC \rightarrow x_2 = \begin{bmatrix} 0.51 \\ 2.25 \end{bmatrix}$$

Solution ① LDA

For two classes the Fisher linear discriminant is

$$W = S_W^{-1}(m_1 - m_2)$$

$$W_0 = -\frac{1}{2} W^T (m_1 + m_2)$$

where $S_W = S_1 + S_2$

the discriminant function is

$$g(x) = W^T x + W_0$$

$$\textcircled{2} S_W = S_1 + S_2$$

$$= \begin{bmatrix} 1042.36 & 54.06 \\ 54.06 & 926.84 \end{bmatrix}$$

$$W = S_W^{-1}(m_1 - m_2)$$

$$= \begin{bmatrix} -0.001265 \\ -0.001997 \end{bmatrix}$$

$$W_0 = -\frac{1}{2} W^T (m_1 + m_2)$$

$$= 0.002970$$

$$\textcircled{4} \text{ For class 1, } g(m_1) = -0.0001569 + 0.002970 > 0$$

If $g(x) > 0$, classifying x to class 1

If $g(x) < 0$, classifying x to class 2

If $g(x)=0$, x on decision boundary

$$\begin{aligned}\textcircled{5} \quad g(x) &= \begin{bmatrix} -0.001265 \\ -0.001997 \end{bmatrix}^T \begin{bmatrix} 1.52 \\ -1.61 \end{bmatrix} + 0.002970 \\ &= 0.001292 + 0.002970 \\ &= 0.004262 > 0\end{aligned}$$

$$g(x_1) > 0$$

So we classify x_1 to class 1

(b) ① Gaussian Naive Bayes

$$p(w_j | x) = \frac{p(x | w_j) p(w_j)}{p(x)} = \frac{\prod_{i=1}^d p(x_i | w_j) p(w_j)}{p(x)}$$

$$p(x | w_j) = \frac{1}{\sqrt{2\pi} \sigma_j} e^{-\frac{1}{2} \left(\frac{x - \mu_j}{\sigma_j} \right)^2}$$

The discriminant function is

$$g_i(x) = \prod_{i=1}^d p(x_i | w_j) p(w_j)$$

for $x_2 = \begin{bmatrix} 0.51 \\ 2.25 \end{bmatrix}$

we denote $x_1 = 0.51$, $x_2 = 2.25$

$$p(w_1) = \frac{600}{1000} = 0.6$$

$$p(w_2) = \frac{400}{1000} = 0.4$$

② calculate Σ_1 and Σ_2

$$\Sigma_1 = \frac{1}{n_1} S_1 = \frac{1}{600} S_1 = \begin{bmatrix} 1.02645 & 0.09305 \\ 0.09305 & 0.8994 \end{bmatrix}$$

$$\Sigma_2 = \frac{1}{n_2} S_2 = \frac{1}{400} S_2 = \begin{bmatrix} 1.066225 & -0.004425 \\ -0.004425 & 0.968 \end{bmatrix}$$

③

we denote σ_{ij}^2 as feature i variance of class w_j

$$\text{So, } \sigma_{11}^2 = 1.02645 \quad \sigma_{21}^2 = 0.8994$$

$$\sigma_{12}^2 = 1.066225 \quad \sigma_{22}^2 = 0.968$$

we denote μ_{ij} as feature i mean of class w_j

$$\text{So, } \mu_{11} = 0.023 \quad \mu_{21} = 0.061$$

$$\mu_{12} = 1.45 \quad \mu_{22} = 1.98$$

$$\begin{aligned} \textcircled{2} p(x_1 | w_1) &= \frac{1}{\sqrt{2\pi} \sigma_{11}} e^{-\frac{1}{2} \left(\frac{x_1 - \mu_{11}}{\sigma_{11}} \right)^2} \\ &= \frac{1}{\sqrt{2\pi} \sqrt{1.02645}} e^{-\frac{1}{2} \left(\frac{0.51 - 0.023}{\sqrt{1.02645}} \right)^2} \\ &= 0.3508 \end{aligned}$$

$$\begin{aligned} p(x_2 | w_1) &= \frac{1}{\sqrt{2\pi} \sigma_{21}} e^{-\frac{1}{2} \left(\frac{x_2 - \mu_{21}}{\sigma_{21}} \right)^2} \\ &= \frac{1}{\sqrt{2\pi} \sqrt{0.8994}} e^{-\frac{1}{2} \left(\frac{2.25 - 0.061}{\sqrt{0.8994}} \right)^2} \\ &= 0.02948 \end{aligned}$$

$$\begin{aligned} p(x_1 | w_2) &= \frac{1}{\sqrt{2\pi} \sigma_{12}} e^{-\frac{1}{2} \left(\frac{x_1 - \mu_{12}}{\sigma_{12}} \right)^2} \\ &= \frac{1}{\sqrt{2\pi} \sqrt{1.06625}} e^{-\frac{1}{2} \left(\frac{0.51 - 1.45}{\sqrt{1.06625}} \right)^2} \\ &= 0.2553 \end{aligned}$$

$$\begin{aligned} p(x_2 | w_2) &= \frac{1}{\sqrt{2\pi} \sigma_{22}} e^{-\frac{1}{2} \left(\frac{x_2 - \mu_{22}}{\sigma_{22}} \right)^2} \\ &= \frac{1}{\sqrt{2\pi} \sqrt{0.968}} e^{-\frac{1}{2} \left(\frac{2.25 - 1.98}{\sqrt{0.968}} \right)^2} \end{aligned}$$

$$= 0.3906$$

$$g_1(x_2) = p(x_1 | w_1) p(x_2 | w_1) p(w_1)$$

$$= 0.3508 \times 0.02948 \times 0.6$$

$$= 0.006205$$

$$g_2(x_2) = p(x_1 | w_2) p(x_2 | w_2) p(w_2)$$

$$= 0.2553 \times 0.3906 \times 0.4$$

$$= 0.03989$$

$$g_2(x_2) > g_1(x_2)$$

So, we classify x_2 to class 2